

INTERNSHIP PROJECT

Text-to-Speech Conversion Web Application

A focused web application that turns written text into spoken audio.

Rohan Sharma · B.Tech – Artificial Intelligence · ABESIT Ghaziabad

PROJECT PREMISE

A simple web app that turns written text into speech

A short path from typed text to playable audio.

01

Web-based Text-to-Speech application

Converts written text into spoken audio.

Uses the ElevenLabs Text-to-Speech API.

02

English and Hindi

Choose the language and voice.

03

Focused internship project

React + Vite frontend with a Node.js/Express backend.

The user journey is short, but complete

01

Input

Enter or paste text.

02

Measure

See character and word counts.

03

Choose

Select language and voice.

04

Generate

Send a validated request for speech.

05

Listen

Play the generated audio or download it.

06

Reset

Clear the form and start again.

Guardrails: input validation and API rate limiting.

The stack separates the interface from the speech engine

Each layer has a focused role, from the React interface to the ElevenLabs API.

LAYER	TECHNOLOGY
Frontend	React + Vite
Styling	Bootstrap + CSS
Backend	Node.js + Express
TTS	ElevenLabs API
Deployment	Vercel + Render
API testing	Postman
Version control	Git + GitHub

Text moves through five clear system stages

01 React frontend

User enters text and chooses language and voice.

02 Express backend

React sends the request to the backend.

03 ElevenLabs TTS API

Express validates the request and calls the external speech engine.

04 Generated MP3

The API returns generated MP3 audio.

05 React player

The frontend loads the audio into a React player.

Three endpoints keep the backend focused

1

GET /api/health

Service health check.

2

POST /api/tts

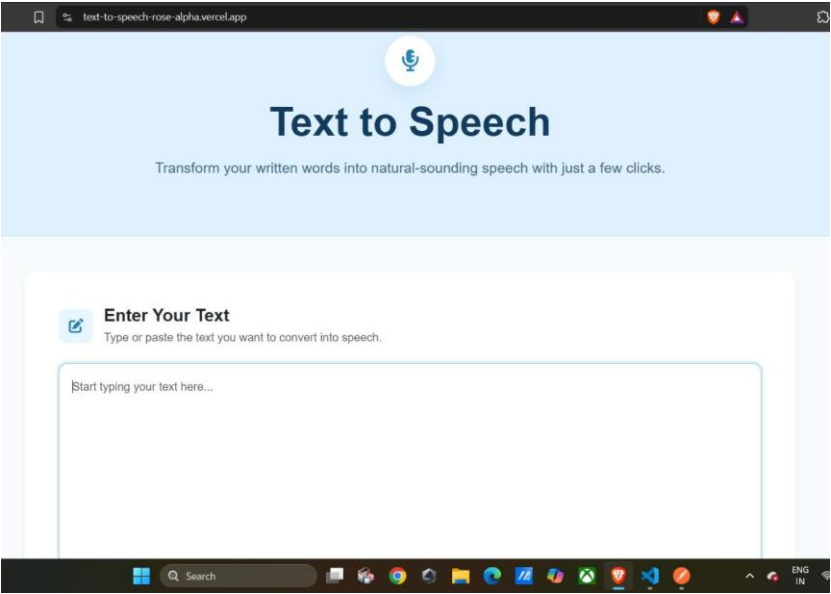
Validate text and request speech generation.

3

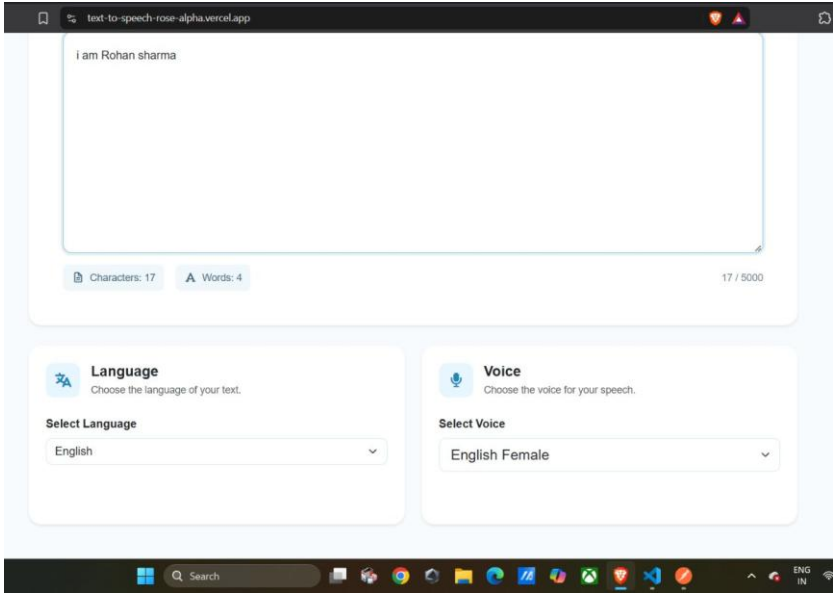
GET /api/voices

Retrieve available voice options.

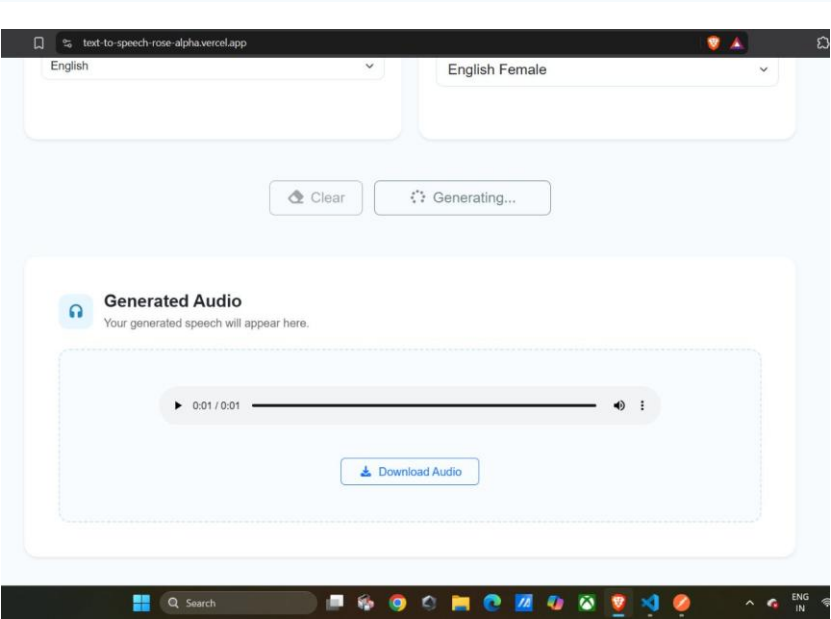
The backend validates the request, communicates with ElevenLabs, receives the generated audio, and returns it to the frontend.



Home



Text input



Generated audio

INTERFACE EVIDENCE

The interface
is built
around one
visible
outcome:
playable
audio

FLOW SUMMARY

Application flow

Text input → Language & voice → Generate →
Audio player

Deployed as a complete, usable project

Deployment

- Frontend: Vercel
- Backend: Render
- Source code: GitHub
- API collection: Postman

Conclusion

The project successfully provides a functional web-based Text-to-Speech system with a React frontend, Express backend, and ElevenLabs API integration.